

# Specifications

## System Hypothesis

- **Asynchronous communication channel**  
using a third-party server to store messages
- **One should be able to communicate with someone without establishing a connection first**
- Participants have not shared public keys at first
- Mallory can read, modify and forge messages send over the network
- **Mallory can read, modify and forge messages on the server**

## Goals

- When two participant exchange a message, the system should protect the confidentiality, integrity and perfect forward secrecy of the messages

GPG - GNU Privacy Guard